

Feature Engineering Logic for Short-Term Load Forecasting

When predicting electricity demand using tabular machine learning models (like LightGBM or XGBoost), the model has no inherent understanding of time, physics, or human behavior. It only sees rows and columns of numbers.

Feature engineering is the process of translating the physical and behavioral realities of the power grid into a language the model can understand. Below is the logical breakdown of the features we create, why they are necessary, and how they conceptually work.

1. Calendar and Human Behavior Features

The Features

- Time Components: Hour of the day, Day of the week, Month of the year.
- Binary Flags: Is_Weekend, Is_Holiday.

Why We Do This

Electricity demand is fundamentally driven by human routines.

- Daily: Commercial load peaks during business hours, while residential load peaks in the early morning and evening.
- Weekly: Factories, schools, and offices drastically reduce their energy consumption on Saturdays and Sundays.
- Annually/Holidays: Specific holidays (like Thanksgiving or Independence Day) have unique load profiles that mimic weekends but with even lower commercial activity.

How We Do It

We extract the basic components from the datetime stamp. We then create binary (0 or 1) flags. If the day is a Saturday or Sunday, Is_Weekend becomes 1. If the date matches a recognized grid holiday, Is_Holiday becomes 1. This explicitly tells the model, "Expect a completely different baseline of demand today."

2. Cyclic Time Encoding (Solving the "Clock" Problem)

The Features

- Sine and Cosine Transformations: Applied to the Hour, Day of the Week, and Month.

Why We Do This

Machine learning models treat numbers linearly. To a model, Hour 23 (11:00 PM) and Hour 0 (Midnight) are 23 units apart. It thinks they are completely unrelated. In reality, they are only 1 hour apart. If the model doesn't understand that time is a circle, it will fail to learn that the load smoothly transitions from the end of one day to the beginning of the next.

How We Do It

We map the linear time values onto a 2D circle using trigonometry (Sine and Cosine). By doing this, the mathematical distance between Hour 23 and Hour 0 becomes very close to zero. We are essentially giving the model a "clock face" so it understands that the end of a cycle seamlessly connects to the beginning of the next.

3. Thermodynamic and Weather Features

The Features

- Heating and Cooling Degree Days (HDD / CDD).
- Squared Temperature.
- Apparent Temperature (Heat Index).

Why We Do This

The relationship between temperature and electricity load is not a straight line; it is a "U-shape".

- When it is very cold, people use massive amounts of energy to heat buildings.
- When it is very hot, people use massive amounts of energy to cool buildings.
- When it is mild (around 65°F / 18°C), neither heating nor cooling is needed, and load drops to its lowest point.

Furthermore, humidity (Dew Point) makes the air feel hotter. Air conditioning units have to work much harder to remove moisture from the air than just lowering the temperature. Finally, buildings have "thermal mass"—they absorb and retain heat, meaning current load is affected by past weather.

How We Do It

- HDD/CDD: We establish a "base comfort temperature" (usually 65°F). If the actual temperature is below 65°F, we calculate the difference to get the Heating Degree. If it's above 65°F, we calculate the difference for the Cooling Degree. This splits the weather impact into two distinct physical drivers.
- Squared Temperature: By squaring the temperature value, we allow the model to easily draw a curved, U-shaped line through the data, rather than forcing it to draw a straight, inaccurate line.
- Apparent Temperature: We combine the dry temperature and the dew point into a single "feels like" metric, capturing the hidden cooling load caused by high humidity.

4. Lag Features (Injecting "Memory")

The Features

- Load Lags: The system load from 1 hour ago, 24 hours ago (yesterday), and 168 hours ago (last week).
- Weather Lags: The temperature and dew point from 1 and 2 hours ago.

Why We Do This

The power grid has immense physical inertia. The most accurate predictor of what the load was right now is what the load was an hour ago. Similarly, human behavior is highly repetitive week-over-week. For weather, because buildings take time to heat up and cool down, the load at 2:00 PM is heavily influenced by the temperature at 12:00 PM, not just 2:00 PM.

How We Do It

We take the historical data and shift it backward in time. We create a new column called Load_1h_Ago and fill it with the actual load from the previous hour. We are manually injecting "memory" into a model that naturally has amnesia, allowing it to see the immediate past context.

5. Rolling Window Features (Capturing Momentum and Volatility)

The Features

- Rolling Averages: The average load and average temperature over the past 3, 6, and 24 hours.
- Rolling Standard Deviations: The volatility/spread of the load over the past 3, 6, and 24 hours.

Why We Do This

A single "Lag" feature (like the load exactly 1 hour ago) might be a temporary anomaly or spike. A rolling average smooths out that noise and tells the model the recent trend or momentum of the grid. The rolling standard deviation tells the model how stable or volatile the grid has been recently. If the standard deviation is high, the model learns to expect sudden swings.

How We Do It

We calculate the mean and the spread of the data over a moving, trailing window of time. Crucially, we ensure the window only looks backward in time so the model is never accidentally given information from the future.

Sources and Foundational Literature

The logic and features outlined above are not arbitrary; they are the established industry standards for Short-Term Load Forecasting (STLF), validated by grid operators, academic research, and global forecasting competitions.

1. Grid Operator & Industry Standards

- ISO New England (ISO-NE) & NERC: The use of a 65°F base temperature for calculating Heating and Cooling Degree Days is the official standard used by ISO-NE in their Regional System Planning and by the North American Electric Reliability Corporation (NERC) for weather-normalizing load data across North America.
- ISO-NE Market Reports: Document the specific, distinct load profiles of recognized market holidays, justifying the creation of binary holiday flags.

2. The "Gold Standard" Competitions (GEFCom)

- Hong, T., et al. (2016). "Probabilistic load forecasting". International Journal of Forecasting. This paper summarizes the methodologies of the winning teams from the Global Energy Forecasting Competition (GEFCom 2014). It proves that tree-based models (like LightGBM/XGBoost) combined with HDD/CDD, temperature lags, and historical load lags consistently outperform complex deep learning models for tabular hourly load data.

3. Academic Literature (Energy Economics & Thermodynamics)

- Pardo, A., Meneu, V., & Valor, E. (2002). "Temperature and day-of-week effects on peak electricity demand". Energy Economics. Mathematically establishes the non-linear, U-shaped relationship between temperature and load, justifying the use of squared temperature terms and splitting weather into HDD/CDD.

- Sailor, D. J., & Munoz, A. A. (1997). *Sensitivity of electricity and natural gas consumption to climate in the USA*. Energy. A foundational paper establishing how Dew Point (humidity) drastically increases summer cooling loads because HVAC systems must work harder to remove latent heat (moisture).
- Amaral, J., et al. (2015). *The impact of temperature and humidity on electricity demand*. Highlights that using "Apparent Temperature" (combining Dry Bulb and Dew Point) yields better load forecasting accuracy than using Dry Bulb alone.

4. Foundational Time-Series & Machine Learning Texts

- Taylor, S. G., & Letham, B. (2018). *Forecasting at scale* (The Prophet paper). *The American Statistician*. Details why time features (hour of day, day of week) must be modeled using Fourier terms (sine/cosine) to capture their cyclical nature, preventing the model from treating the end and beginning of a cycle as distant outliers.
- Box, G. E. P., Jenkins, G. M., et al. *Time Series Analysis: Forecasting and Control*. The foundational text for time-series analysis. Establishes the principle of autocorrelation (that current values are heavily dependent on past values), which is the theoretical basis for using Lags and Rolling Windows in machine learning.